

12.5) (1)

$$\begin{cases} V_{GS} = -0.33 I_D \\ I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_{GS(off)}} \right)^2 \end{cases}$$

$$I_{DSS} = 9 \text{ mA} \quad V_{GS(off)} = -3 \text{ V}$$

$$I_D = I_{DSS} \left(1 - \frac{-0.33 I_D}{-3} \right)^2$$

$$= I_{DSS} (1 - 0.11 I_D)^2$$

$$= I_{DSS} (1 + 0.0121 I_D^2 - 0.22 I_D)$$

$$= 9 (1 + 0.0121 I_D^2 - 0.22 I_D)$$

$$I_D = 9 + 0.1089 I_D^2 - 1.98 I_D$$

$$0.1089 I_D^2 - 2.98 I_D + 9 = 0$$

$$I_D \approx 3.457 \text{ mA} \quad V_{GS} \approx -1.141 \text{ V}$$

$$V_{DS} = V_{DD} - I_D R_D - I_S R_S$$

$$= 9 - 3.457 \times 10^{-3} (1000) - 3.457 \times 10^{-3} (330)$$

$$= 4.402 \text{ V}$$

$$(2) \quad A_u = -g_m R_L' = -g_m (R_D // R_L) = -3.7 \times (10 // 10) = -3.3636 \quad r_i = R_G = 10 \text{ M}\Omega$$

$$U_{out}(\text{rms}) = |A_u| U_{in} = 3.3636 (10) = 33.636 \text{ mV}$$

$$(12.7)(1) V_G = \frac{R_2}{R_1 + R_2} \cdot V_{DD} = \frac{70}{170}(20) = 8.2353V$$

$$V_{GS} = V_G - V_S = 8.2353 - 0.5 R_S = 3.2353V$$

$$A_u = -g_m R'_L = -g_m (R_D // R_L) = -0.5 \left(\frac{1}{R_D} + \frac{1}{10} \right)^{-1} = -2.5$$

$$V_{DS} = V_{DD} - I_D R_D - I_D R_S$$

$$10 = 20 - (0.5)(10) - (0.5)R_S$$

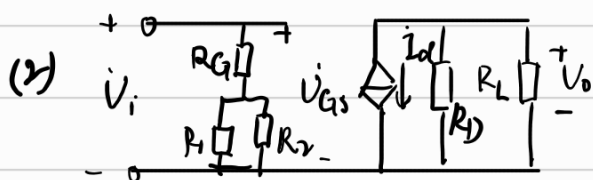
$$R_S = 10k\Omega$$

$$10R_D / (10 + R_D) = 5$$

$$10R_D = 50 + 5R_D$$

$$5R_D = 50$$

$$R_D = 10k\Omega$$



$$(3) r_i = R_G + (R_1 // R_2) = 1000 + \left(\frac{1}{\frac{1}{100} + \frac{1}{70}} \right)^{-1} = 1.0412M\Omega$$

$$r_o = R_D = 10k\Omega$$

$$(4) A_u = -2.5 = \frac{\dot{U}_o}{\dot{U}_i} \quad \text{设 } \dot{U}_i = 0.2 \angle 0^\circ V$$

$$\dot{U}_o = -0.5 = 0.5 \angle 180^\circ V \quad u_o = 0.5 \sin(\omega t + 180^\circ) V$$

$$(5) \dot{U}_o = |A_u| \frac{r_i}{R_S + r_i} \dot{U}_i = 2.5 \left(\frac{1.0412}{1.0412 + 0.01} \right) (0.2) = 0.4952V$$

$$u_o = 0.4952 \sin \omega t V$$

$$12.9) (1) V_{B1} \approx V_{CC} \cdot \frac{R_{12}}{R_{11} + R_{12}} = \frac{4}{1+1} \left(\frac{18}{3+18} \right) = 4V$$

$$I_{E1} = \frac{V_{B1} - V_{BE1}}{R_{E1}} = \frac{4 - 0.7}{3} = 1.1 \text{ mA}$$

$$r_{be1} = 300 + (1 + \beta_1) \left(\frac{26}{I_{E1}} \right) = 300 + 51 \left(\frac{26}{1.1} \right) = 1505.4545 \approx 1.5055 \text{ k}\Omega$$

$$V_{B2} \approx V_{CC} \cdot \left(\frac{10}{30} \right) = 4V$$

$$I_{E2} = \frac{V_{B2} - V_{BE2}}{R_{E2}} = \frac{4 - 0.7}{2} = 1.65 \text{ mA}$$

$$r_{be2} = 300 + 51 \left(\frac{26}{1.65} \right) = 1.1036 \text{ k}\Omega$$

$$r_{i2} = R_{21} // R_{22} // r_{be2} = 0.9469 \text{ k}\Omega$$

$$A_{u1} = \frac{U_{o1}}{U_{i1}} = -\beta_1 \frac{R_{c1} // r_{i2}}{r_{be1}} = -80 \left(\frac{0.7197}{1.5055} \right) = -23.9025$$

$$A_{u2} = \frac{U_{o2}}{U_{i2}} = -\beta_2 \frac{R_{c2} // R_L}{r_{be2}} = -80 \left(\frac{2.8}{1.1036} \right) = -113.2657$$

$$A_u = A_{u1} \cdot A_{u2} = 2707.3328$$

$$U_o = A_u U_i = 2707.3328 (0.5) \sin \omega t \text{ mV} = 1353.6664 \sin \omega t \text{ mV}$$

$$(2) r_i = R_{11} // R_{12} // r_{be1} \approx 1.3085 \text{ k}\Omega$$

$$r_o = R_{E2} = 5 \text{ k}\Omega$$



$$(2) \quad V_B \approx 12 \cdot \left(\frac{5.6}{45.6} \right) = 1.4737 \text{ V} \quad I_E = \frac{V_B - V_{BE}}{R_E} = \frac{1.4737 - 0.7}{1.2} = 0.6448 \text{ mA}$$

$$r_{be} = 300 + 61 \left(\frac{26}{0.6448} \right) = 2.7597 \text{ k}\Omega$$

$$A_{u1} = -g_m (R_D // r_{i2}) = -1 (5 // 40 // 56 // 2.7597) = -1.3056$$

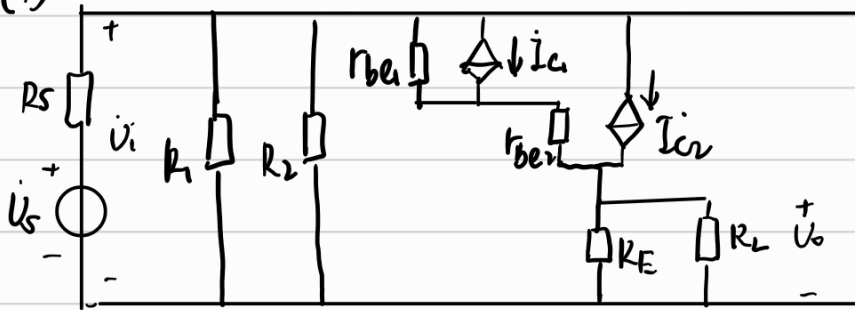
$$A_{u2} = -\beta \left(\frac{R_C // R_L}{r_{be}} \right) = -60 \left(\frac{5 // 5}{2.7597} \right) = -54.3537$$

$$A_u = A_{u1} \cdot A_{u2} = 70.9642$$

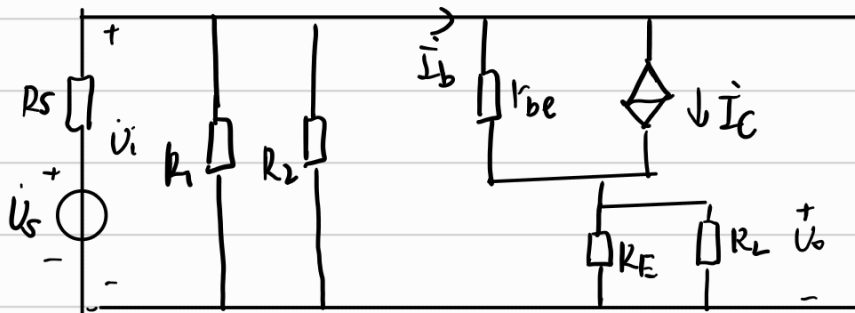
$$r_i = R_G + R_{i1} // R_{i2} = 265.3846 \text{ k}\Omega$$

$$R_o = R_C = 5 \text{ k}\Omega$$

12.11) (1)



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(2)

$$r_{be} = r_{be1} + (1 + \beta_1) r_{be2} \quad I_C = I_{C1} + I_{C2} = \beta_1 I_B + \beta_2 (1 + \beta_1) I_B$$

$$= [\beta_1 + \beta_2 (1 + \beta_1)] I_B$$

$$I_E = I_B + I_C = [\beta_1 + \beta_2 (1 + \beta_1) + 1] I_B$$

$$I_B \approx \frac{V_{CC} - U_{BE}}{R_1 // R_2 + (\beta + 1) R_E} \approx \frac{15 - 1.4}{120 + 2800}$$

$$\beta \approx \beta_1 \beta_2 = 2800$$

$$R_E' = R_E // R_L = 5 k\Omega$$

$$\approx 0.4847 \mu A$$

$$I_E = [70 + 40(71) + 1] 0.4847 \times 10^{-6}$$

$$= 1.411 mA$$

$$r_{be} = 300 + (\beta + 1) \left(\frac{26}{I_E} \right) = 300 + 2801 \left(\frac{26}{1.411} \right) = 51.913 k\Omega$$

$$U_i = I_{B1} r_{be1} + I_{B1} (1 + \beta_1) r_{be2} + I_{B1} (1 + \beta_1) (1 + \beta_2) R_E'$$

$$= [r_{be1} + (1 + \beta_1) r_{be2} + (1 + \beta_1) (1 + \beta_2) R_E'] I_{B1}$$

$$= [r_{be} + (1 + \beta_1) (1 + \beta_2) R_E'] I_{B1}$$

$$\dot{I}_i = \dot{I}_{e1} + \dot{I}_{e2} + \dot{I}_{b1} = \frac{U_i}{R_1} + \frac{U_i}{R_2} + \frac{U_i}{r_{be1} + (1+\beta_1)r_{be2} + (1+\beta_1)(1+\beta_2)R_L'}$$

$$\begin{aligned} r_i = \frac{U_i}{\dot{I}_i} &= R_1 // R_2 // r_{be1} + (1+\beta_1)r_{be2} + (1+\beta_1)(1+\beta_2)R_L' \\ &\approx R_1 // R_2 // r_{be} + (1+\beta_1)(1+\beta_2)R_L' \\ &\approx 119.0222 \text{ k}\Omega \end{aligned}$$

$$A_{us} = \frac{U_o}{U_s} = A_u \frac{r_i}{R_s + r_i} = 0.9963 \left(\frac{119.0222}{10 + 119.0222} \right) = 0.9191$$

$$A_u = \frac{(\beta+1)R_L'}{r_{be} + (\beta+1)R_L'} = \frac{2801(5)}{51.913 + 289(5)} = 0.9963$$